CIVIL ENGINEERING & STRUCTURAL DESIGN

TASK 2: STRUCTURAL LOAD ANALYSIS

CodeAlpha Internship

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Task	Structural Load Analysis — simple reinforced-concrete beam
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Objective

Estimate the loads acting on a simple reinforced-concrete beam and calculate its factored load, support shear, maximum bending moment, and an approximate elastic deflection under service load. This is a basic educational example.

1. Given data and assumptions

Parameter	Assumed value	Note
Beam span	4.0 m	Simply supported
Beam section	300 mm × 500 mm	Width × overall depth
Slab tributary width	3.0 m	Load transferred to beam
Slab thickness	0.15 m	Concrete unit weight 25 kN/m³
Finishes/services	1.0 kN/m²	Illustrative allowance
Floor live load	2.0 kN/m²	Illustrative only; code must confirm
Concrete modulus E	25 GPa	Simplified deflection estimate

2. Dead-load calculation

Slab self-weight = thickness × unit weight = 0.15 × 25 = 3.75 kN/m²

Total area dead load = slab self-weight + finishes = 3.75 + 1.00 = 4.75 kN/m²

Line load from slab = 4.75 kN/m² × 3.0 m = 14.25 kN/m

Beam self-weight = 0.30 × 0.50 × 25 = 3.75 kN/m

Total dead line load D = 14.25 + 3.75 = 18.00 kN/m

3. Live load and factored design load

Live line load L = 2.0 kN/m² × 3.0 m = 6.00 kN/m

For this example only, use the illustrative ultimate combination 1.2D + 1.6L. The governing code may require different combinations.

w■ = 1.2(18.00) + 1.6(6.00) = 21.60 + 9.60 = 31.20 kN/m

Service load = D + L = 18.00 + 6.00 = 24.00 kN/m

4. Shear force and bending moment

For a simply supported beam with a uniformly distributed load over the full span:

Maximum support shear: V■■■ = w■L / 2 = 31.20 × 4.0 / 2 = 62.40 kN

Maximum mid-span bending moment: M■■■ = w■L² / 8 = 31.20 × 4.0² / 8 = 62.40 kN·m

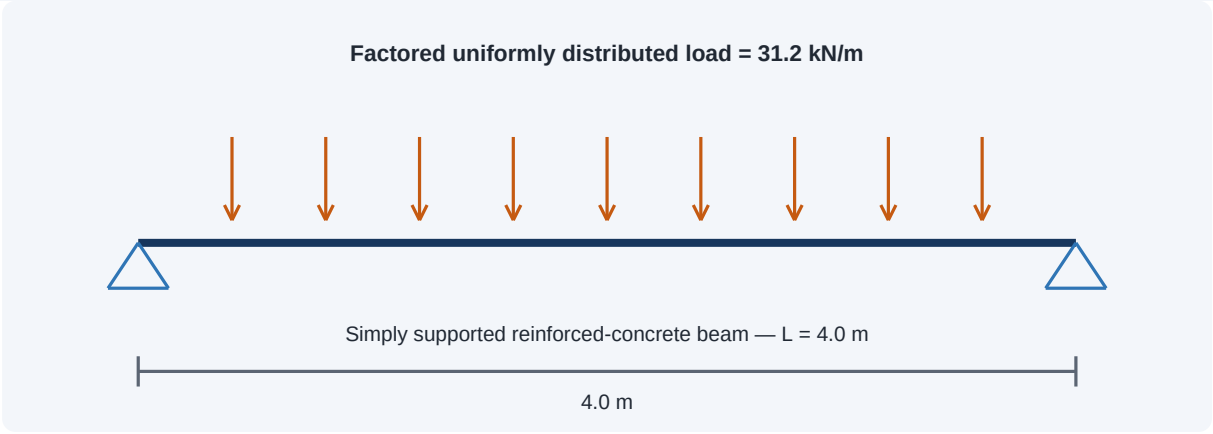

Figure 1. Simply supported beam under a uniform factored load.

5. Approximate elastic deflection

For a simply supported beam under uniform load, δ■■■ = 5wL■/(384EI). Use service load w = 24,000 N/m, L = 4 m, E = 25 × 10■ N/m².

Second moment of area: I = bh³/12 = 0.30 × 0.50³ / 12 = 0.003125 m■

δ■■■ ≈ 5(24,000)(4■) / [384(25×10■)(0.003125)] = 0.00256 m = 2.56 mm

This idealized estimate ignores concrete cracking, reinforcement details, creep, shrinkage, and long-term effects. It is not a final code deflection check.

6. Results and structural behavior

Item	Calculated result	Interpretation
Dead line load D	18.00 kN/m	Permanent load
Live line load L	6.00 kN/m	Variable occupancy load
Factored load w■	31.20 kN/m	Illustrative combination
Maximum support shear	62.40 kN	At the supports
Maximum bending moment	62.40 kN·m	At mid-span
Approx. deflection	2.56 mm	Idealized service estimate

Structural behavior

- The beam bends downward under gravity loads; the maximum positive bending moment is at mid-span for this loading case.
- The two support reactions are equal due to the symmetrical span and load. Each reaction is 62.40 kN.
- Shear force is greatest near the supports and reduces to zero at mid-span in the idealized case.
- A real design must check flexural strength, shear strength, reinforcement detailing, anchorage, cracking, deflection, durability, and fire resistance.

7. Limitations

This report estimates the actions on a beam; it does not design reinforcement or certify structural safety. A full design needs the governing local code, confirmed material strengths, reinforcement details, actual tributary width, wall loads, load combinations, seismic/wind requirements where applicable, and qualified engineering review. Do not use these numbers directly for construction.

8. Conclusion

This example demonstrates how slab and beam self-weight are converted to a line load, how live load is added, and how basic shear, bending moment, and deflection estimates are calculated. These are important first steps before a complete code-based design.

Design basis: Basic structural mechanics equations for a simply supported beam under a uniformly distributed load. All dimensions and loads in this example are assumed for learning and must be verified against applicable local codes.